

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Atmakuri Sai Mrudula (192372087) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Sai Mrudula

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

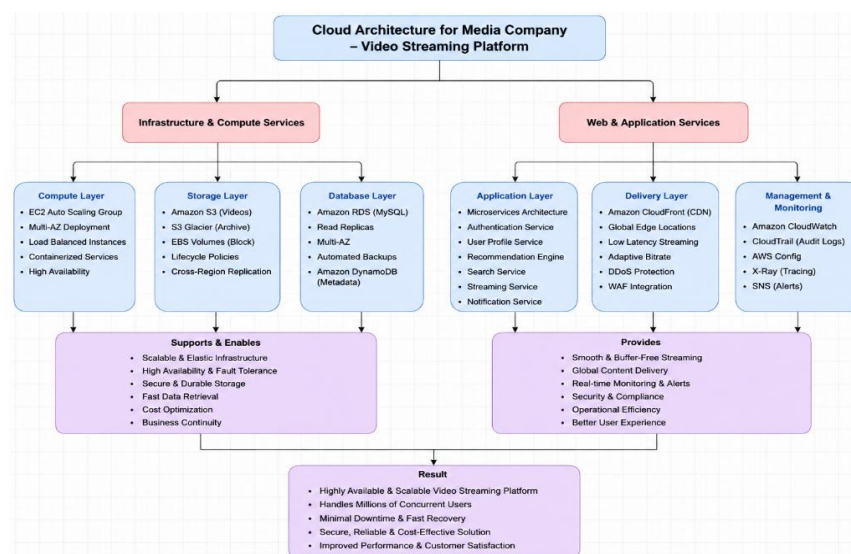
Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

With the rapid growth of digital entertainment, millions of users watch movies, TV shows, and live events simultaneously through online streaming platforms. During popular events, user traffic can increase dramatically within a short period. Traditional on-premises infrastructure often struggles to handle these sudden spikes, leading to slow response times and video buffering. Cloud computing provides an ideal solution because it offers scalability, flexibility, and high availability. By using cloud services, the media company can automatically increase computing resources when demand rises and reduce them when demand decreases, ensuring uninterrupted streaming while controlling infrastructure costs.

Cloud Architecture Design: -



The proposed cloud architecture begins with users accessing the streaming platform through the internet. Their requests are first directed to a Content Delivery Network (CDN), which stores copies of popular videos at edge locations close to users. This significantly reduces latency because users receive content from the nearest server instead of the main data center.

The application servers handle user authentication, subscription validation, video recommendations, and streaming requests. Video files are stored in highly durable cloud object storage, while user profiles, watch history, payment details, and subscription information are maintained in managed databases. A distributed caching system such as Redis stores frequently accessed information in memory, allowing data to be retrieved much faster than from the database.

Scalability Techniques: -

Scalability is one of the most important characteristics of cloud computing. It enables applications to handle increasing workloads without affecting performance. In this architecture, horizontal scaling is primarily used. Instead of upgrading a single server, additional application servers are created automatically whenever user traffic increases. This approach allows the platform to support millions of concurrent users efficiently.

The application is developed using a microservices architecture, where each function such as authentication, streaming, payment processing, recommendations, and notifications operates independently. Since every service can be scaled separately, resources are allocated more efficiently.

Caching Techniques: -

Caching improves performance by storing frequently requested data in temporary high-speed memory. Instead of retrieving information repeatedly from the database or storage system, the application serves cached data almost instantly.

The CDN caches video files at geographically distributed edge servers, reducing bandwidth usage and providing faster content delivery. Application-level caching stores user sessions, authentication tokens, and recommendation results. Database caching stores frequently executed queries, reducing database workload.

Load Balancing and Auto Scaling: -

Load balancing ensures that incoming requests are evenly distributed across multiple servers. This prevents performance degradation caused by overloaded servers. If one server becomes unavailable, the load balancer automatically redirects traffic to healthy servers, improving fault tolerance and service availability.

Auto Scaling continuously monitors system performance. When CPU utilization, memory usage, or network traffic exceeds predefined thresholds, additional application servers are launched automatically. During periods of low demand, unnecessary servers are removed to reduce infrastructure costs. This elastic resource allocation allows the platform to maintain excellent performance while avoiding overprovisioning.

Cost Optimization Strategies: -

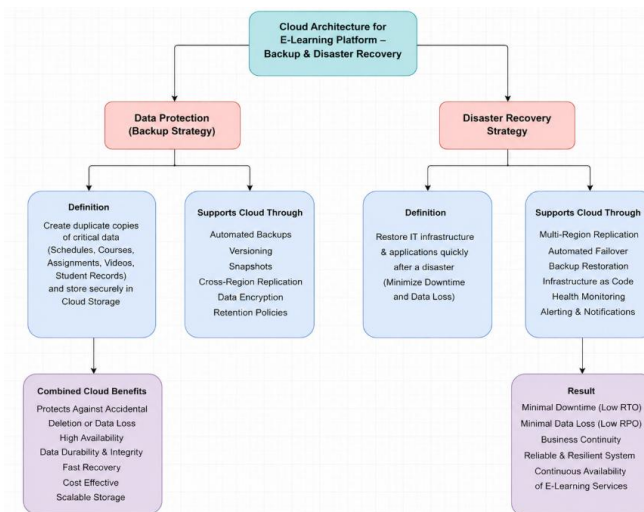
- Although cloud platforms provide unlimited scalability, proper cost management is essential. The organization can reduce operational expenses by implementing several optimization strategies.
- Using a CDN decreases bandwidth consumption because cached content is served from nearby edge servers. Storage lifecycle policies automatically move older videos to lower-cost archival storage.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

The financial technology industry relies heavily on cloud computing to deliver secure, reliable, and high-performance digital banking, online payment, investment, and financial management services. As the number of customers increases, applications must be deployed quickly, updated frequently, and remain available without interruption. To achieve these goals, many organizations use container technology and multi-cloud deployment.

Proposed Cloud Architecture: -

The proposed solution uses Kubernetes clusters deployed across multiple cloud providers. Applications are first packaged into Docker containers. These container images are stored in a centralized Container Registry, making them accessible to every cloud environment. Kubernetes clusters running in AWS, Azure, and Google Cloud deploy identical containerized applications, ensuring consistency across all platforms.



A centralized monitoring system continuously collects logs, performance metrics, and health information from every Kubernetes cluster. Service Mesh technologies such as Istio provide secure communication between microservices and simplify traffic management across cloud providers. This architecture ensures high availability, automatic scaling, and efficient workload distribution.

Container Orchestration Using Kubernetes: -

Kubernetes acts as the central management platform for all containerized applications. Whenever developers release a new version of an application, Kubernetes automatically deploys updated containers without causing service interruption. If any container fails unexpectedly, Kubernetes immediately detects the failure and restarts the affected container without requiring manual intervention.

Kubernetes also distributes containers across multiple worker nodes, ensuring that application workloads are balanced efficiently. If one node becomes unavailable due to hardware failure or maintenance, Kubernetes automatically schedules affected containers on healthy nodes, maintaining continuous service availability.

Multi-Cloud Strategy: -

A multi-cloud strategy allows the organization to distribute workloads among different cloud providers based on performance, pricing, compliance, or geographic location. Critical banking applications can run simultaneously in multiple cloud environments, ensuring uninterrupted services even if one cloud provider experiences an outage.

Sensitive financial information can be stored in specific regions to satisfy government regulations, while customer-facing applications can be deployed closer to users for improved response times. Traffic can be intelligently routed to the cloud provider offering the lowest latency or highest availability. This approach not only improves user experience but also prevents vendor lock-in because applications remain portable across different cloud platforms.

Benefits of Multi-Cloud Deployment: -

1. Deploying applications across multiple cloud providers offers several important advantages.
2. The organization is no longer dependent on a single cloud provider, eliminating vendor lock-in. If one cloud provider experiences an outage, workloads automatically continue operating in another cloud, improving business continuity.
3. Multiple cloud regions also improve disaster recovery because applications remain available even during regional failures.
4. Organizations can select cloud providers based on cost, performance, compliance requirements, or available services. This flexibility enables better workload optimization and improved customer experience across different geographic locations.
5. Multi-cloud deployment also supports regulatory compliance by allowing sensitive financial information to remain within specific countries while still delivering global application services.

Risks and Challenges: -

1. Although multi-cloud architectures provide numerous advantages, they also introduce several challenges.
2. Managing infrastructure across multiple cloud providers is considerably more complex than managing a single cloud environment. Each provider offers different networking models, security configurations, monitoring tools, and billing systems.
3. Data synchronization between cloud providers may introduce additional latency. Transferring data between clouds can also increase operational costs due to network bandwidth charges.
4. Maintaining consistent security policies across multiple environments requires careful planning. Identity management, encryption, access control, and compliance monitoring must be standardized to avoid security vulnerabilities.
5. Monitoring distributed applications also becomes more difficult because logs and performance metrics originate from multiple cloud platforms.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

Cloud computing has become an essential technology for modern e-learning platforms because it provides scalability, flexibility, and easy access to educational resources from anywhere in the world. These platforms store a large amount of important data, including student records, assignments, examination results, lecture videos, course materials, and user information. Since this data is valuable, accidental deletion, hardware failures, cyberattacks, or natural disasters can result in significant losses and disrupt learning activities.

Cloud-Native Backup and Disaster Recovery Architecture: -

The proposed solution uses multiple cloud-native services working together to protect educational data. Student records, assignments, videos, and course materials are stored in highly durable cloud object storage. Automatic versioning is enabled so that every modification creates a new copy instead of replacing the existing file. If a file is accidentally deleted or modified, previous versions can be restored immediately.

Databases containing student information and examination results are backed up automatically using scheduled snapshots. Continuous database replication copies changes to another cloud region, ensuring that recent data remains available even if the primary region fails.

Cloud-Native Backup Strategy: -

A successful backup strategy should include different types of backups.

- Full Backup creates a complete copy of all application data. It provides maximum protection but requires more storage space and longer backup time.
- Incremental Backup stores only the changes made since the previous backup. This method reduces storage requirements and backup duration.
- Differential Backup stores all changes made since the last full backup, providing a balance between recovery speed and storage efficiency.

Data Replication: - Replication is the process of maintaining multiple copies of data in different locations.

Synchronous Replication: - Writes data simultaneously to multiple storage locations before confirming the transaction. This provides excellent consistency but may introduce slight delays because both copies must be updated at the same time.

Asynchronous Replication: - First writes data to the primary location and then copies it to secondary locations. Although there may be a small delay, this approach provides better performance and is commonly used for cross-region disaster recovery.

Object Versioning: -

Versioning is one of the most effective cloud-native features for preventing accidental data loss. Whenever a user edits or deletes a file, the previous version is automatically preserved. Administrators can restore any earlier version without affecting other files. For educational institutions, versioning is especially useful because course materials and assignments are frequently updated throughout the academic year.

Recovery Time Objective (RTO): -

Recovery Time Objective (RTO) represents the maximum acceptable amount of downtime after a disaster. For an e-learning platform, long periods of downtime may interrupt online classes, examinations, and assignment submissions. Therefore, RTO should be as short as possible.

Low RTO can be achieved through:

- Standby disaster recovery infrastructure.
- Automated failover systems.
- Fast snapshot restoration.
- Infrastructure as Code for rapid deployment.
- Continuous monitoring and automated alerts.

Recovery Point Objective (RPO): -

Recovery Point Objective (RPO) defines the maximum acceptable amount of data loss after a failure. For example, if the RPO is five minutes, the organization can tolerate losing only the last five minutes of data.

Low RPO is achieved using:

- Continuous database replication.
- Frequent automated backups.
- Transaction log shipping.
- Real-time synchronization.
- Regular snapshot creation.

Business Continuity During Failures: -

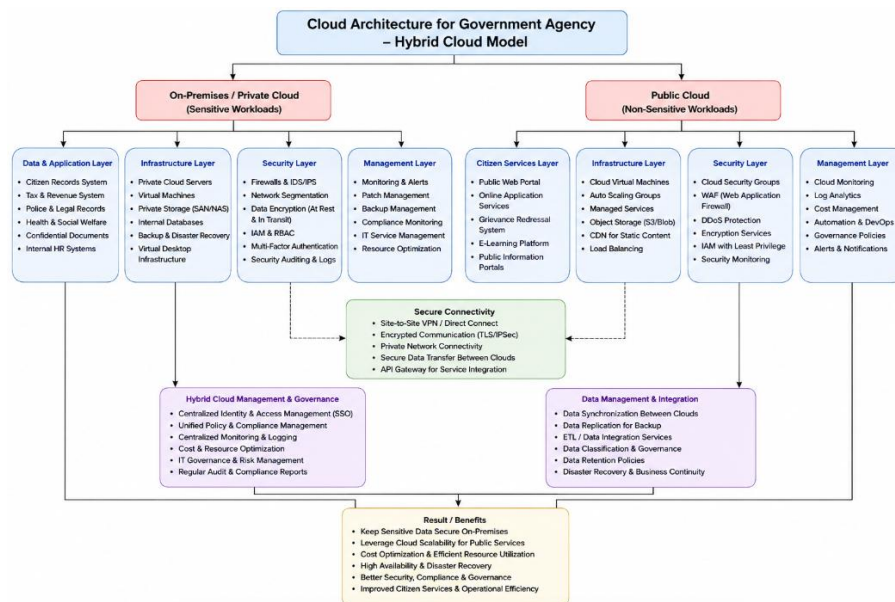
- Business continuity ensures that educational services continue even during unexpected failures.
- The organization should first detect the failure using monitoring systems. Automated alerts immediately notify administrators whenever backup failures or application outages occur.
- Disaster recovery procedures are then activated automatically. User traffic is redirected to the standby cloud region using DNS failover or load balancing. Data is restored from replicated storage or recent backups, and application services are restarted.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

A Hybrid Cloud combines the advantages of both private (on-premises) and public cloud environments. Sensitive applications remain inside secure government data centers, while non-sensitive workloads are deployed on the public cloud. This approach allows organizations to maintain security and regulatory compliance while benefiting from the flexibility, scalability, and cost-effectiveness of cloud computing.

Hybrid Cloud Architecture Design: -

The proposed architecture separates workloads according to their sensitivity. Highly confidential applications, including citizen databases, tax systems, police records, and defense-related applications, remain inside the government's on-premises data center or private cloud.



Secure Communication Between Environments: -

Secure communication is essential because sensitive information frequently moves between on-premises systems and cloud applications. The organization should establish encrypted communication channels using IPsec VPN or dedicated private connectivity services. These secure connections protect data from interception while it travels across public networks.

Identity Management: -

Identity Management ensures that only authorized users can access government resources. A centralized Identity and Access Management (IAM) system should manage all users across both on-premises and cloud environments. Employees should authenticate through a single identity provider using Single Sign-On (SSO), allowing secure access to multiple applications without repeatedly entering credentials.

Role-Based Access Control (RBAC) assigns permissions based on employee responsibilities. For example, healthcare officers should only access medical records, while finance officers should access financial systems.

Multi-Factor Authentication (MFA) should be implemented for all privileged users. In addition to passwords, users must provide another authentication factor such as a mobile verification code or biometric authentication.

Encryption Strategy: -

Encryption protects confidential government information from unauthorized access. Data stored inside databases, file systems, and cloud storage should use AES-256 encryption, which is one of the strongest encryption standards currently available. Data transmitted between cloud services and on-premises infrastructure should use TLS 1.3 encryption to prevent attackers from intercepting sensitive information.

Managing Hybrid Cloud Environments: -

Managing a hybrid cloud infrastructure requires careful coordination between cloud services and on-premises systems. Infrastructure monitoring should provide a unified view of servers, applications, storage systems, databases, and network devices. Automated configuration management tools ensure that software versions and security policies remain consistent across all environments. Infrastructure as Code tools simplify deployment by automating server provisioning and configuration. Regular software updates, vulnerability assessments, and security audits reduce operational risks. Disaster recovery plans should include both cloud and on-premises resources to ensure rapid service restoration during emergencies.

Challenges in Hybrid Cloud Management: -

1. Network latency may occur when applications continuously exchange large volumes of data between environments.
2. Maintaining consistent security policies across multiple platforms requires significant planning and administrative effort.
3. Data synchronization between cloud databases and local databases can become complicated, especially during high transaction volumes.

Benefits of Hybrid Cloud: -

1. Organizations can reduce operational costs because only critical applications require expensive private infrastructure.
2. Hybrid cloud also improves disaster recovery by allowing backup systems to operate in cloud environments.
3. Scalability becomes much easier because additional cloud resources can be allocated automatically whenever demand increases.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Software as a Service (SaaS) has become one of the most widely used cloud computing models because it allows users to access software applications over the internet without installing or maintaining them locally. Organizations use SaaS applications for customer relationship management, enterprise resource planning, collaboration, online learning, and many other business functions. Since users expect these applications to be available at all times, even a few minutes of downtime can lead to financial losses, reduced customer satisfaction, and damage to the company's reputation.

Multi-Region Cloud Architecture: -

The proposed architecture deploys identical copies of the SaaS application in multiple cloud regions. Users access the application through the internet using a Global DNS Service. User requests are first directed to a Global Load Balancer, which automatically routes traffic to the nearest healthy cloud region based on latency and availability.

Multi-Region Deployment Strategy: -

The SaaS provider can implement one of two deployment models.

1. Active-Active Deployment: -

In the Active-Active model, all cloud regions actively serve customer requests at the same time. Traffic is distributed among multiple regions using Global Load Balancers. If one region becomes unavailable, traffic is automatically redirected to the remaining healthy regions without interrupting user services.

2. Active-Passive Deployment: -

In the Active-Passive model, one region operates as the primary production environment while another region remains on standby. The standby region continuously receives replicated data but does not normally process customer requests. If the primary region fails, automated failover activates the standby region and redirects user traffic.

Data Replication Strategy: -

Data replication ensures that information remains available in multiple locations. Within the same cloud region, Synchronous Replication copies every database transaction to multiple database servers before confirming the transaction. This guarantees strong consistency and prevents data loss if one database server fails.

Between different cloud regions, Asynchronous Replication copies data continuously with minimal performance impact. Although there may be a slight delay, this approach provides excellent disaster recovery capabilities.

Failover Mechanism: -

1. Failover is the process of automatically switching application services from a failed region to a healthy region.
2. Monitoring systems continuously perform health checks on application servers, databases, and networking components.
3. If the monitoring system detects that a region has become unavailable, the Global Load Balancer immediately stops sending traffic to the failed region and redirects users to another operational region.
4. Database replicas are automatically promoted to primary databases whenever necessary, allowing applications to continue processing user requests without manual intervention.
5. Once the failed region becomes operational again, it is synchronized with the active region before rejoining the load balancing pool.

Monitoring and Alerting Strategy: -

Continuous monitoring is essential for maintaining application reliability. Cloud monitoring tools collect information about CPU utilization, memory usage, storage capacity, network throughput, database performance, application response time, and error rates. Health monitoring also verifies that replication processes, load balancers, virtual machines, containers, and cloud services are functioning correctly.

Predictive monitoring uses Artificial Intelligence and Machine Learning to identify abnormal behavior before failures occur. Early detection allows administrators to resolve issues proactively and prevent service disruptions. Centralized dashboards provide real-time visibility into the health of every cloud region, enabling faster troubleshooting and operational decision-making.

Ensuring High Availability: -

1. High availability is achieved through several cloud technologies working together.
2. Load balancing distributes incoming requests evenly among healthy servers.
3. Auto Scaling automatically launches additional application instances whenever workload increases.
4. Multiple Availability Zones provide redundancy within each cloud region.
5. Cross-region deployment protects against complete regional failures.

Benefits of the Proposed Solution: -

1. The proposed multi-region architecture provides numerous benefits for SaaS providers.
2. It significantly improves application availability by eliminating dependence on a single cloud region.
3. Customers experience lower latency because traffic is directed to the nearest geographical location.
4. Automatic failover reduces service interruptions during unexpected failures.
5. Data replication protects valuable customer information from loss.